BTS-VI(R/S)-05-24-2485 Reg. No. 20621067

B

B.Tech. Degree VI Semester Regular/Supplementary Examination May 2024

SE 19-206-0607 (IE) SAFETY IN PETROLEUM AND PETROCHEMICAL INDUSTRIES

(2019 Scheme)

Time: 3 Hours Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

CO1: Interpret the various processes employed in petroleum refining and petrochemical industries.
CO2: Estimate the firewater requirement for a petroleum refinery and petroleum depots.
CO3: Recognize the hazards involved in on-shore and off-shore drilling and state control measures.
CO4: Describe the various sources of ignition in process industries and the fire prevention systems employed.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 - Create

PO – Programme Outcome

PART A

(Answer *ALL* questions)

		(8 × 3 = 24)	Marks	BL	CO	PO
I.	(a)	Summarize on the ethane recovery process in petroleum industries.	3	L2	1	1
	(b)	Explain the process of hydro forming.	3	L2	1	2
	(c)	Discuss the different types of storage tanks.	3	L1	2	3
	(d)	Write a note on the salient features of OISD 116 and 117.	3	L2	2	1
	(e)	Discuss the requirements of standby pumps as per any one OISD.	3	L2	3	2
	(f)	Summarize the construction requirements for cross country hydrocarbon pipelines.	3	L2	3	2
	(g)	Discuss the methods of rescue and fire fighting in an offshore platform for crude oil recovery.	3	L1	4	1
	(h)	Enlist the classification of wells for fossil fuel recovery based on various criterions.	3	L2	4	3

PART B

		(4 × 12 = 48)				
II.		With the help of neat diagram, elucidate on the process of separation of various components from crude oil.	12	L2	1	1
		OR				
III.		Explain the process of sulfur recovery and butadiene extraction with the help of flow diagrams.	12	L2	1	3
IV.		Summarize the design concepts of various fixed fire protection systems used in petroleum and petrochemical industries.	12	L3	2	2
		OR				
V.	(a)	Briefly discuss the significance of tank layout in preventing accidents in a petroleum depot.	7	L3	2	2
	(b)	Define a tank farm. Briefly explain the key features of tank farm. Discuss about the fire protection measures provided in a tank farm.	5	L2	2	2

(P.T.O.)

			Marks	BL	CO	PO
VI.	(a)	Discuss the design philosophy of LPG bottling plant as per the relevant OISD guideline.	6	L2	3	3
	(b)	Discuss on the safety and fire protection systems in bottling plants. Also enlist the safety measures to be followed in the storage of bulk LPG.	6	L3	3	2

OR

			Marks	BL	CO	PO
VII.		Estimate the fire water requirement for a petroleum depot with following units. Assume a justifiable value for missing data if any.	12	L4	3	3,4

Description	No. of Tanks	Height (m)	Diameter(m)	No. of tanks inside R+30
Floating Roof tank	3	12	75	1
Cone Roof tank	1	16	32	0
LPG sphere	2	-	16	-

			Marks	BL	CO	PO
VIII.		Enumerate on the process of crude oil production using rotary drilling process with the help of a neat diagram.	12	L3	4	3

OR

			Marks	BL	CO	PO
IX.	(a)	Enlist the various platforms used for offshore drilling. Explain the advantages and disadvantages of each platform.	6	L2	4	3
	(b)	How can you access an underground oil or gas reserve by drilling in a non-vertical direction? Explain its features.	6	L3	4	2

Blooms's Taxonomy Level's
L1- 5%; L2 - 49.17%; L3 - 35.83%; L4 - 10%.

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